

Phase 1: Core Writing & Archive — Detailed Specification

Table of Contents

1. Overview
2. User Stories
3. Information Architecture
4. Frontend Specification
5. Backend Specification
6. Database Design
7. API Contract
8. State Management
9. Error Handling & Edge Cases
10. Performance Considerations
11. Security Requirements
12. Testing Strategy
13. Deployment Checklist
14. Post-Launch Monitoring

1. Overview

Purpose

Enable the fundamental creative loop: a user reads a canonical text, rewrites it, and publishes that rewrite to a searchable archive. This is the atomic unit of PlotOri. Every subsequent feature—peer review, badges, classroom tools, anthologies—presupposes this loop functioning flawlessly.

Scope Boundaries

- **In scope:** User registration, canonical text browsing, drafting canvas with auto-save, submission publishing, public archive with search/filter
- **Explicitly out of scope:** Peer review, rubric evaluation, weekly workflows, badges, admin moderation beyond text upload, student zone, multi-lingual support, AI/plagiarism detection, intertextual tools, temporal transposition, social features (follow, like, comment)

Duration

6–7 weeks (assuming 1–2 full-time developers)

Success Criteria

A first-time visitor can complete the full creative loop—browse a text, register, draft a reimagining, and publish it—in under 10 minutes without confusion or errors.

2. User Stories

Story 1: Discovery

As a literature enthusiast, I want to browse canonical texts on PlotOri so that I can find a work I know well and feel inspired to rewrite.

Acceptance Criteria:

- Landing page loads in under 2 seconds
- I can see at least 5 canonical texts with cover imagery or visual identifiers
- Each text card shows title, author, year, and a one-line genre tag
- Clicking a text takes me to its detail page within 1 second
- The detail page shows a curated passage (300–800 words) that invites rewriting

Story 2: Registration

As a visitor, I want to create an account quickly so that I can save my drafts and publish my work.

Acceptance Criteria:

- Registration form requires only email, username, and password
- Username is unique and 3–30 characters (alphanumeric + underscore)
- Email is validated for format and uniqueness
- Password minimum 8 characters
- OAuth option available (Google, GitHub)
- Account is active immediately (no email verification required for Phase 1)
- I am redirected to the text I was browsing before registering

Story 3: Drafting

As a registered user, I want to write my reimagining alongside the original text so that I can stay faithful to the source while being creative.

Acceptance Criteria:

- Clicking “Reimagine This” opens the Drafting Canvas
- Original text is visible on the left (or top on mobile), my draft on the right (or bottom)
- I can scroll the original text independently of my draft
- The editor supports paragraphs, italics, and dialogue formatting
- Auto-save triggers every 30 seconds of active typing
- A “Last saved” timestamp updates visibly
- If I accidentally close the tab, my draft is recoverable on return
- I can save as draft or proceed to publish

Story 4: Publishing

As a drafter, I want to finalize and publish my reimagining so that it becomes part of the public archive.

Acceptance Criteria:

- Pre-publish review screen shows my full draft
- I must provide a title (max 100 characters)
- I must select an activity type: “Alternative Ending” or “Counterfactual Writing”
- I must write an author note (50–300 characters) explaining my creative intent
- I can optionally add theme tags (up to 5)
- Publishing is irreversible (no unpublish in Phase 1)
- On publish, I am redirected to my submission’s public page

Story 5: Archive Browsing

As any visitor, I want to browse published reimaginings so that I can read others’ creative work and find inspiration.

Acceptance Criteria:

- Archive shows all published submissions in reverse chronological order
- I can filter by source text, activity type, or date range
- I can search by title, author username, or tag
- Results paginate at 20 items per page
- Each card shows title, source text, author, type, and first 150 characters of content
- Clicking a card opens the full submission page

3. Information Architecture

Site Map

```
/
├─ / (Landing Page)
│   ├── Hero section with mission statement
│   ├── Featured submissions (latest 3)
│   ├── Browse canonical texts (grid of 5-7 cards)
│   └── CTA to register or explore archive
├─ /texts
│   ├── /texts/{canonical_text_id}
│   │   ├── Text metadata header
│   │   ├── Curated source passage (expandable)
│   │   ├── Existing submissions linked to this text
│   │   └── "Reimagine This" button (requires login)
```

```

|
├─ /write/{canonical_text_id}
|   └─ Drafting Canvas (protected route)
|       ├── Left panel: Source text (read-only, scrollable)
|       └─ Right panel: Editor (auto-saving)
|
├─ /submit/{draft_id}
|   └─ Pre-publish review form (protected route)
|       ├── Title input
|       ├── Activity type selector
|       ├── Author note textarea
|       ├── Theme tag selector
|       └─ Publish button
|
├─ /archive
|   └─ Filterable, searchable grid of submissions
|       ├── /archive?text={id}&type={type}&date={range}
|       └─ /archive?search={query}
|
├─ /submission/{submission_id}
|   └─ Public submission page
|       ├── Title and metadata
|       ├── Full reimagined text
|       ├── Author profile link
|       └─ Source text link
|
├─ /profile/{username}
|   └─ Public profile
|       ├── Bio and avatar
|       └─ List of published works
|
├─ /register
|   └─ Registration form
|
├─ /login
|   └─ Login form
|
└─ /me
    └─ Personal dashboard (protected)
        ├── Drafts in progress
        └─ Published works

```

4. Frontend Specification

4.1 Landing Page (/)

Layout:

- Navigation bar: Logo, Browse Texts, Archive, Login/Register (or Profile if authenticated)

- Hero section: Full-width, mission statement in large type, subheading explaining the creative loop
- Featured submissions: 3 cards showing the most recent high-quality submissions (seed content initially)
- Canonical texts grid: 5–7 cards in a responsive grid (3 columns desktop, 2 tablet, 1 mobile)
- Footer: About, Contact, Terms, Privacy

States:

- Loading: Skeleton cards for texts and submissions
- Empty: If no featured submissions, show seed content or a “Be the first to reimagine” prompt
- Error: Retry button if texts fail to load

4.2 Text Detail Page (/texts/{id})

Layout:

- Breadcrumb: Home > Texts > [Title]
- Header: Title (large), author, year, genre badge, era badge
- Source passage: Collapsible section, default expanded, max-height with scroll for long passages
- “Reimagine This” button: Primary CTA, disabled with tooltip if not logged in
- Linked submissions: Horizontal scroll or grid of cards showing existing reimaginings of this text

Interactions:

- Source passage can be collapsed/expanded
- “Reimagine This” redirects to /write/{id} if logged in, or /login?redirect=/texts/{id} if not

4.3 Drafting Canvas (/write/{canonical_text_id})

Layout:

- Full-screen or near-full-screen layout
- Header bar: Text title (clickable, opens in new tab), save status, word count, “Review & Publish” button
- Main area: Two-panel layout (60/40 or 50/50 split)
- Left: Source text panel (read-only, syntax-highlighted or styled to distinguish from draft)
- Right: Editor panel (rich text input)
- Footer bar: Auto-save status indicator, “Save Draft” manual button, keyboard shortcut hints

Responsive Behavior:

- Desktop (>1024px): Side-by-side panels
- Tablet (768–1024px): Side-by-side with narrower panels
- Mobile (<768px): Stacked—source text as collapsible top section, editor below

Editor Behavior:

- Rich text controls: Bold, italic, paragraph, blockquote (for dialogue)
- No font/size/color controls—maintain platform consistency
- Tab key inserts 4 spaces (for dialogue indentation)
- Auto-save: Debounced at 30 seconds of typing inactivity, plus immediate on manual “Save Draft”
- Recovery: On page load, check for unsaved localStorage draft; if found and newer than server version, prompt user to restore

State Machine:

```
[LOADING] → Fetch source text and any existing draft
  ↓
[READY] → User can type
  ↓ (typing detected)
[DIRTY] → Auto-save timer starts
  ↓ (30s inactivity)
[SAVING] → API call to save draft
  ↓ (success)
[SAVED] → Timestamp updated, state returns to READY
  ↓ (error)
[ERROR] → Show retry option, keep DIRTY state
```

4.4 Pre-Publish Review (/submit/{draft_id})

Layout:

- Header: “Review Your Reimagining”
- Two-column preview: Left shows original text excerpt, right shows your draft (read-only)
- Form fields:
 - Title: Text input, character counter (0/100)
 - Activity Type: Radio buttons (Alternative Ending, Counterfactual Writing)
 - Author Note: Textarea, character counter with min/max (50/300)
 - Theme Tags: Multi-select dropdown or tag input (max 5)
- Actions: “Back to Edit” (returns to canvas), “Save as Draft” (returns to dashboard), “Publish” (irreversible)

Validation:

- Title: Required, 3–100 characters
- Activity Type: Required
- Author Note: Required, 50–300 characters
- Tags: Optional, max 5

Confirmation:

- Modal on “Publish” click: “This cannot be undone. Your work will be public.”
- Success: Redirect to /submission/{new_id} with toast notification

4.5 Archive Page (/archive)

Layout:

- Header: “Archive” with result count
- Filter bar (sticky on scroll):
- Source text: Dropdown of all canonical texts
- Activity type: Dropdown
- Date range: Preset buttons (Today, This Week, This Month, All Time) or custom
- Search: Text input with magnifying glass icon
- Results grid: 20 cards per page
- Pagination: Numbered buttons + Previous/Next

Card Design:

- Title (bold, truncated if long)
- Source text title (smaller, muted)
- Author username with avatar
- Activity type badge
- Date published (relative: “2 hours ago”, “3 days ago”)
- First 150 characters of content (faded, ellipsis if truncated)

Empty States:

- No results: “No reimaginings found. Try adjusting filters or be the first to write one.”
- No submissions at all: “The archive is empty. Start writing to fill it.”

4.6 Submission Page (/submission/{id})

Layout:

- Header: Title, author profile link, date published
- Metadata row: Source text link, activity type badge, theme tags
- Main content: Full reimagined text in readable typography (max-width 680px centered, 18px font, 1.6 line-height)
- Author note: Italicized block below content, labeled “Author’s Note”
- Source reference: “Based on [Text Title] by [Author]” at bottom

Interactions:

- Clicking source text title navigates to /texts/{id}
- Clicking author navigates to /profile/{username}

4.7 Profile Page (/profile/{username})

Layout:

- Header: Avatar (generated or uploaded), display name, username, bio
- Stats row: Member since, total submissions
- Tabs: Published Works | Drafts (only visible to owner)
- Published Works: Grid same as archive cards

4.8 Personal Dashboard (/me)

Layout:

- Header: “Your Dashboard”
 - Drafts section: List of in-progress drafts with last edited date, word count, “Continue Writing” button
 - Published section: List of published works with stats (views, date)
-

5. Backend Specification

5.1 Authentication Service

Registration:

- Endpoint: POST /api/auth/register
- Body: { email, username, password }
- Validation:
- Email: Valid format, not already registered
- Username: 3–30 chars, alphanumeric + underscore, unique
- Password: Minimum 8 characters
- Response: { user_id, username, token }
- Error cases: Duplicate email, duplicate username, weak password

Login:

- Endpoint: POST /api/auth/login
- Body: { email, password } or { username, password }
- Response: { user_id, username, token }
- Error cases: Invalid credentials, account not found

OAuth (Optional for Phase 1):

- Endpoints: GET /api/auth/google, GET /api/auth/github
- Callback: GET /api/auth/{provider}/callback
- Flow: Standard OAuth 2.0 authorization code flow

- On first login, create account with provider email and generated username

Token Management:

- JWT access tokens, 24-hour expiry
- Refresh tokens, 7-day expiry
- Token refresh endpoint: POST /api/auth/refresh

5.2 Canonical Text Service

List Texts:

- Endpoint: GET /api/texts
- Response: Array of text objects with metadata
- Query params: ?genre=&era=&author= for filtering
- Sort: Alphabetical by title, default

Get Single Text:

- Endpoint: GET /api/texts/{id}
- Response: Full text object including source_passage
- Cache: 1-hour server-side cache (texts are read-heavy)

Admin: Create Text (Manual/Seed Only):

- Endpoint: POST /api/admin/texts
- Protected: Admin role only (hardcoded admin list for Phase 1)
- Body: { title, author, year, genre, era, summary, source_passage }

5.3 Submission Service

Create Draft:

- Endpoint: POST /api/drafts
- Body: { canonical_text_id, title, content, activity_type }
- Response: { draft_id, created_at }
- Rules: User must be authenticated, canonical_text_id must exist

Auto-Save Draft:

- Endpoint: PATCH /api/drafts/{draft_id}
- Body: { content, title? }
- Response: { saved_at, version_id }
- Behavior: Creates new version in submission_versions table
- Rate limit: Max 1 save per 10 seconds per draft (prevent spam)

Get Draft:

- Endpoint: GET /api/drafts/{draft_id}

- Response: Full draft with latest content and all versions
- Authorization: Must be draft owner

List My Drafts:

- Endpoint: GET /api/me/drafts
- Response: Array of drafts with metadata (title, canonical_text_title, last_edited, word_count)

Publish Submission:

- Endpoint: POST /api/submissions
- Body: { draft_id, title, activity_type, author_note, tag_ids? }
- Validation:
- Draft must exist and belong to user
- Title: 3–100 chars
- Activity type: Must be valid enum
- Author note: 50–300 chars
- Side effects:
- Creates submission record
- Copies latest draft content to submission
- Sets published_at timestamp
- Deletes draft record (or marks as converted)
- Associates tags
- Response: { submission_id, published_at, url }

Get Submission:

- Endpoint: GET /api/submissions/{submission_id}
- Response: Full submission with author profile, canonical text metadata, tags
- Public: No authentication required
- Cache: 5-minute server-side cache

List Submissions (Archive):

- Endpoint: GET /api/submissions
- Query params:
- ?text_id= — filter by canonical text
- ?activity_type= — filter by type
- ?date_from=&date_to= — date range
- ?search= — full-text search on title and author_note
- ?page= — pagination (default 1)

- ?per_page= — default 20, max 50
- Response: { submissions: [...], total_count, page, per_page, total_pages }
- Sort: published_at descending

5.4 Tag Service

List Tags:

- Endpoint: GET /api/tags
- Response: Array of { id, name, category }
- Categories: theme, tone, device

6. Database Design

6.1 Entity Relationship Diagram

```
users ||--o{ submissions : writes
users ||--o{ drafts : owns
canonical_texts ||--o{ submissions : inspires
canonical_texts ||--o{ drafts : inspires
submissions ||--o{ submission_tags : has
tags ||--o{ submission_tags : belongs_to
drafts ||--o{ draft_versions : saves
```

6.2 Table Specifications

users

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email
username	VARCHAR(30)	UNIQUE, NOT NULL	Display handle
display_name	VARCHAR(50)	NULL	Optional full name
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hash
bio	TEXT	NULL	Max 500 chars
avatar_url	VARCHAR(255)	NULL	Generated or uploaded
is_admin	BOOLEAN	DEFAULT FALSE	Hardcoded admin flag
created_at	TIMESTAMP	DEFAULT NOW()	Account creation
updated_at	TIMESTAMP	DEFAULT NOW()	Last profile update

canonical_texts

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
title	VARCHAR(255)	NOT NULL	Work title
author	VARCHAR(255)	NOT NULL	Author name

Table 2 – continued

Column	Type	Constraints	Description
year	INTEGER	NULL	Publication year
genre	VARCHAR(100)	NOT NULL	Tragedy, Novel, etc.
era	VARCHAR(100)	NULL	Elizabethan, Victorian, etc.
summary	TEXT	NOT NULL	200–500 char overview
source_passage	TEXT	NOT NULL	300–800 word excerpt
added_by	UUID	FK → users.id	Admin who added
created_at	TIMESTAMP	DEFAULT NOW()	When added to platform

drafts

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
user_id	UUID	FK → users.id, NOT NULL	Owner
canonical_text_id	UUID	FK → canonical_texts.id, NOT NULL	Source text
title	VARCHAR(100)	NULL	Working title
content	TEXT	NOT NULL	Latest draft content
activity_type	VARCHAR(50)	NULL	Enum if selected
word_count	INTEGER	DEFAULT 0	Computed on save
last_saved_at	TIMESTAMP	DEFAULT NOW()	Last auto-save
created_at	TIMESTAMP	DEFAULT NOW()	Draft creation

draft_versions

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
draft_id	UUID	FK → drafts.id, NOT NULL, CASCADE DELETE	Parent draft
content	TEXT	NOT NULL	Snapshot content
saved_at	TIMESTAMP	DEFAULT NOW()	When this version saved

submissions

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
user_id	UUID	FK → users.id, NOT NULL	Author

Table 5 – continued

Column	Type	Constraints	Description
canonical_text_id	UUID	FK → canonical_texts.id, NOT NULL	Source text
title	VARCHAR(100)	NOT NULL	Published title
content	TEXT	NOT NULL	Final published content
author_note	VARCHAR(300)	NOT NULL	Creative intent explanation
activity_type	VARCHAR(50)	NOT NULL	Alternative Ending or Counterfactual
published_at	TIMESTAMP	NOT NULL	Publication time
created_at	TIMESTAMP	DEFAULT NOW()	Record creation

tags

Column	Type	Constraints	Description
id	UUID	PK, auto-gen	Unique identifier
name	VARCHAR(50)	UNIQUE, NOT NULL	Tag label
category	VARCHAR(20)	NOT NULL	theme, tone, or device

submission_tags

Column	Type	Constraints	Description
submission_id	UUID	FK → submissions.id, CASCADE DELETE	
tag_id	UUID	FK → tags.id, CASCADE DELETE	
PK	(submission_id, tag_id)	Composite PK	

6.3 Indexes

Table	Columns	Type	Purpose
users	email	B-tree	Fast login lookup
users	username	B-tree	Fast profile lookup
submissions	user_id	B-tree	Author' s works list
submissions	canonical_text_id	B-tree	Text-linked submissions
submissions	published_at	B-tree	Archive sorting
submissions	activity_type	B-tree	Type filtering
draft_versions	draft_id, saved_at	B-tree	Version history retrieval

6.4 Seed Data

Canonical Texts (5 initial):

Title	Author	Year	Genre	Era	Summary	Source Passage
Othello	William Shakespeare	1604	Tragedy	Elizabethan	A Moorish general in Venice is manipulated by his ensign Iago into believing his wife Desdemona is unfaithful, leading to tragic consequences.	[Act 3, Scene 3: Othello's temptation speech through Act 4 Scene 1: The handkerchief confrontation — approximately 600 words]
Macbeth	William Shakespeare	1606	Tragedy	Elizabethan	A Scottish general receives a prophecy that he will become king, leading him and his wife to murder and madness in their pursuit of power.	[Act 1, Scene 7: Macbeth's dagger soliloquy through Act 2 Scene 2: The murder aftermath — approximately 650 words]
Pride and Prejudice	Jane Austen	1813	Novel	Regency	The romantic clash between Elizabeth Bennet and Mr. Darcy explores themes of manners, upbringing, morality, education, and marriage in Georgian England.	[Chapter 34: Darcy's first proposal and Elizabeth's rejection — approximately 700 words]

Table 9 – continued

Title	Author	Year	Genre	Era	Summary	Source Passage
Jane Eyre	Charlotte Brontë	1847	Novel	Victorian	An orphaned governess falls in love with her employer Mr. Rochester, only to discover a terrible secret hidden in the attic of Thornfield Hall.	[Chapter 27: Rochester' s confession and Jane' s flight from Thornfield – approximately 750 words]
Great Expectations	Charles Dickens	1861	Novel	Victorian	An orphan named Pip rises through society with the help of an unknown benefactor, learning painful lessons about wealth, love, and personal integrity.	[Chapter 38: Pip confronts Miss Havisham about Estella' s manipulation – approximately 600 words]

Tags (Initial Set):

Name	Category
Hopeful	tone
Tragic	tone
Ironic	tone
Ambiguous	tone
Restorative	tone
Love	theme
Betrayal	theme
Power	theme
Identity	theme
Class	theme
Gender	theme
Metaphor	device
Foreshadowing	device

Table 10 – continued

Name	Category
Irony	device

7. API Contract

7.1 Authentication

POST /api/auth/register Request:

```
{
  "email": "reader@example.com",
  "username": "literary_rewriter",
  "password": "securePass123"
}
```

Success 201:

```
{
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "username": "literary_rewriter",
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expires_at": "2026-05-28T00:00:00Z"
}
```

Error 409 (Duplicate):

```
{
  "error": "Conflict",
  "message": "Username already taken",
  "field": "username"
}
```

POST /api/auth/login Request:

```
{
  "email": "reader@example.com",
  "password": "securePass123"
}
```

Success 200: Same as register response

Error 401:

```
{
  "error": "Unauthorized",
  "message": "Invalid email or password"
}
```

7.2 Canonical Texts

GET /api/texts Success 200:

```
{
  "texts": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440001",
      "title": "Othello",
      "author": "William Shakespeare",
      "year": 1604,
      "genre": "Tragedy",
      "era": "Elizabethan",
      "summary": "A Moorish general in Venice...",
      "submission_count": 12
    }
  ],
  "total": 5
}
```

GET /api/texts/{id} Success 200:

```
{
  "id": "550e8400-e29b-41d4-a716-446655440001",
  "title": "Othello",
  "author": "William Shakespeare",
  "year": 1604,
  "genre": "Tragedy",
  "era": "Elizabethan",
  "summary": "A Moorish general in Venice...",
  "source_passage": "Othello: O, now, for ever... [full passage]",
  "submission_count": 12,
  "recent_submissions": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440010",
      "title": "What If Emilia Spoke Sooner",
      "author": "classic_rewriter",
      "published_at": "2026-05-20T14:30:00Z"
    }
  ]
}
```

7.3 Drafts

POST /api/drafts Headers: Authorization: Bearer {token}

Request:

```
{
  "canonical_text_id": "550e8400-e29b-41d4-a716-446655440001",
  "title": "What If Emilia Spoke Sooner",
  "content": "Emilia: My husband? Othello: What of him?...",
}
```

```
"activity_type": "Alternative Ending"
}
```

Success 201:

```
{
  "draft_id": "550e8400-e29b-41d4-a716-446655440020",
  "created_at": "2026-05-26T10:00:00Z",
  "word_count": 450
}
```

PATCH /api/drafts/{draft_id} Headers: Authorization: Bearer {token}

Request:

```
{
  "content": "Emilia: My husband? Othello: What of him?... [updated]",
  "title": "What If Emilia Exposed Iago"
}
```

Success 200:

```
{
  "saved_at": "2026-05-26T10:30:00Z",
  "version_id": "550e8400-e29b-41d4-a716-446655440021",
  "word_count": 520
}
```

Error 429 (Rate Limit):

```
{
  "error": "Too Many Requests",
  "message": "Save allowed every 10 seconds",
  "retry_after": 7
}
```

GET /api/drafts/{draft_id} Success 200:

```
{
  "id": "550e8400-e29b-41d4-a716-446655440020",
  "canonical_text": {
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "title": "Othello",
    "author": "William Shakespeare",
    "source_passage": "..."
  },
  "title": "What If Emilia Exposed Iago",
  "content": "Current draft content...",
  "activity_type": "Alternative Ending",
  "word_count": 520,
  "last_saved_at": "2026-05-26T10:30:00Z",
  "versions": [
```

```
{
  "version_id": "550e8400-e29b-41d4-a716-446655440021",
  "saved_at": "2026-05-26T10:30:00Z",
  "word_count": 520
},
{
  "version_id": "550e8400-e29b-41d4-a716-446655440022",
  "saved_at": "2026-05-26T10:15:00Z",
  "word_count": 480
}
]
```

7.4 Submissions

POST /api/submissions Headers: Authorization: Bearer {token}

Request:

```
{
  "draft_id": "550e8400-e29b-41d4-a716-446655440020",
  "title": "What If Emilia Exposed Iago",
  "activity_type": "Alternative Ending",
  "author_note": "I wanted to explore how Desdemona's fate changes if Emilia breaks her silence before
  ⇨ the murder, testing whether truth can outmaneuver Iago's poison.",
  "tag_ids": ["550e8400-e29b-41d4-a716-446655440100", "550e8400-e29b-41d4-a716-446655440101"]
}
```

Success 201:

```
{
  "submission_id": "550e8400-e29b-41d4-a716-446655440030",
  "published_at": "2026-05-26T11:00:00Z",
  "url": "/submission/550e8400-e29b-41d4-a716-446655440030"
}
```

Error 400 (Validation):

```
{
  "error": "Bad Request",
  "message": "Validation failed",
  "details": {
    "author_note": "Must be between 50 and 300 characters"
  }
}
```

GET /api/submissions Query params: ?text_id=&activity_type=&date_from=&date_to=&search=&page=1&per_page=

Success 200:

```
{
  "submissions": [
    {
```

```

    "id": "550e8400-e29b-41d4-a716-446655440030",
    "title": "What If Emilia Exposed Iago",
    "author": {
      "username": "literary_rewriter",
      "display_name": "Literary Rewriter",
      "avatar_url": "https://..."
    },
    "canonical_text": {
      "id": "550e8400-e29b-41d4-a716-446655440001",
      "title": "Othello",
      "author": "William Shakespeare"
    },
    "activity_type": "Alternative Ending",
    "published_at": "2026-05-26T11:00:00Z",
    "preview": "Emilia: My husband? Othello: What of him? Emilia: He is a villain, my lord. He stole
    ⇨ my handkerchief and...",
    "tags": ["Tragic", "Betrayal"]
  }
],
"total_count": 47,
"page": 1,
"per_page": 20,
"total_pages": 3
}

```

GET /api/submissions/{id} Success 200:

```

{
  "id": "550e8400-e29b-41d4-a716-446655440030",
  "title": "What If Emilia Exposed Iago",
  "content": "Full reimaged text here...",
  "author_note": "I wanted to explore how Desdemona's fate changes...",
  "activity_type": "Alternative Ending",
  "published_at": "2026-05-26T11:00:00Z",
  "author": {
    "username": "literary_rewriter",
    "display_name": "Literary Rewriter",
    "bio": "Rewriting the classics one scene at a time.",
    "avatar_url": "https://...",
    "member_since": "2026-05-25T08:00:00Z"
  },
  "canonical_text": {
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "title": "Othello",
    "author": "William Shakespeare",
    "year": 1604,
    "source_passage": "Original passage here..."
  },
  "tags": [
    { "id": "...", "name": "Tragic", "category": "tone" },
    { "id": "...", "name": "Betrayal", "category": "theme" }
  ]
}

```

```
  ]  
}
```

8. State Management

8.1 Frontend State

Global State (Context/Store):

- **auth:** Current user, token, login/logout actions
- **ui:** Toast notifications, modal states, loading indicators

Local/Page State:

- **textBrowser:** Texts list, filters, pagination
- **draftingCanvas:** Draft content, auto-save status, word count, original text
- **submissionReview:** Form values, validation errors
- **archive:** Submissions list, filters, search query, pagination

8.2 Draft Persistence Strategy

Three-Layer Persistence:

1. **In-Memory (React State):** Active draft content lives here during editing
2. **localStorage (Browser):** Every keystroke debounced to localStorage with key `plotori_draft_{draft_id}`
3. **Server (Database):** Auto-saved every 30 seconds of inactivity

Recovery Flow on Page Load:

```
Check server for latest draft version  
↓  
Check localStorage for `plotori_draft_{draft_id}`  
↓  
Compare timestamps  
↓  
If localStorage newer → Show recovery prompt: "You have unsaved changes from [time]. Restore?"  
If server newer → Load server version, clear localStorage  
If equal → Load normally
```

Cleanup:

- On successful publish: Clear localStorage key
 - On explicit "Discard Draft": Clear localStorage and delete server draft
 - On logout: Clear all localStorage draft keys
-

9. Error Handling & Edge Cases

9.1 Network Errors

Scenario	User Experience	Technical Action
Auto-save fails	Toast: “Changes not saved. Retrying…”	Exponential backoff retry (2s, 4s, 8s), max 3 attempts
Auto-save repeatedly fails	Banner: “Connection unstable. Your work is safe locally.”	localStorage continues, manual save button highlighted
Publish fails	Modal: “Unable to publish. Keep this tab open and try again.”	Draft remains intact, user can retry
Archive fails to load	Skeleton with retry button	Cache previous results, show stale data with “Updated X ago”

9.2 Concurrent Editing

Scenario: User opens draft in two tabs. **Resolution:**

- Last-write-wins at the version level
- On save, check if server version changed since load
- If conflict: Show modal with diff, let user choose which version to keep

9.3 Lost Connection During Drafting

Scenario: User loses WiFi for 30 minutes. **Resolution:**

- localStorage accumulates all changes
- On reconnect, detect large time gap
- Prompt: “You have [X] unsaved changes from offline. Sync now?”
- Sync creates new version, preserves offline work

9.4 Browser Crash

Scenario: Browser crashes with unsaved draft. **Resolution:**

- localStorage survives browser restart
- On next visit to `/write/{id}`, detect recovery data
- Show prominent recovery banner

9.5 Accidental Navigation

Scenario: User clicks browser back button with unsaved changes. **Resolution:**

- `beforeunload` event listener shows native confirmation dialog
- `onbeforeunload = (e) => { if (isDirty) e.preventDefault(); }`

10. Performance Considerations

10.1 Load Time Budgets

Page	Target	Maximum
Landing	1.5s	2.5s
Text Detail	1.0s	2.0s
Drafting Canvas	2.0s	3.5s
Archive	1.5s	2.5s
Submission Page	1.0s	2.0s

10.2 Optimization Strategies

Frontend:

- Code-split by route (lazy load archive, drafting canvas)
- Preload canonical texts data on landing page hover
- Debounce search input at 300ms
- Virtualize archive list if >100 items rendered
- Compress draft content with LZ-string before localStorage (5MB limit)

Backend:

- Cache canonical texts for 1 hour (rarely change)
- Cache submission pages for 5 minutes (read-heavy)
- Database connection pooling (default 10 connections)
- Query optimization: Archive queries use covering indexes
- Rate limiting: 100 requests/minute per IP, 10 saves/minute per user

Database:

- Archive pagination at database level, not application
- Full-text search using PostgreSQL `tsvector` (not `LIKE` queries)
- Connection pooling via PgBouncer if scaling beyond 100 concurrent users

10.3 Bundle Size Budgets

Chunk	Target	Maximum
Landing + Core	150KB	250KB
Drafting Canvas	100KB	180KB
Archive	80KB	150KB
Vendor (shared)	200KB	350KB

11. Security Requirements

11.1 Authentication

- Passwords hashed with bcrypt (cost factor 12)
- JWT signed with RS256 (asymmetric, private key on server only)
- Tokens transmitted in Authorization header, never cookies (prevent CSRF)
- Refresh tokens stored hashed in database

11.2 Authorization

- Draft endpoints: Verify `draft.user_id === request.user_id`
- Submission endpoints: Verify ownership or public access
- Admin endpoints: Check `users.is_admin === true`
- No direct object reference vulnerabilities (validate all UUIDs belong to requesting user)

11.3 Input Validation

- All text inputs sanitized (strip HTML except allowed formatting tags)
- Content Security Policy: `default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'`
- Rate limiting on registration (5 attempts per IP per hour)
- Maximum content length: 50,000 characters per draft (prevent abuse)

11.4 Data Protection

- No PII beyond email and username
 - Email visible only to user and admin
 - Drafts are private until published
 - No tracking cookies or third-party analytics in Phase 1
-

12. Testing Strategy

12.1 Unit Tests

Backend:

- Authentication: Registration validation, login flow, token generation
- Draft service: CRUD operations, version creation, ownership checks
- Submission service: Publish validation, tag association, archive queries
- Rate limiting: Threshold enforcement, reset timing

Frontend:

- Drafting Canvas: Auto-save debouncing, word count accuracy, localStorage sync
- Form validation: Title length, author note bounds, tag limits
- Archive filters: URL param synchronization, pagination logic

12.2 Integration Tests

Critical Paths:

1. Register → Browse text → Start draft → Auto-save → Publish → View in archive
2. Login → Continue draft from localStorage recovery → Publish
3. Browse archive → Filter by text → Filter by type → Search → View submission

Error Scenarios:

- Auto-save failure with retry
- Concurrent tab editing conflict
- Invalid publish attempt (missing fields)
- Unauthorized draft access

12.3 End-to-End Tests

Using Playwright or Cypress:

- Complete user journey: Anonymous visitor → registered user → published author
- Mobile responsive: Drafting canvas on 375px width
- Performance: Archive loads with 50 submissions in <2s

12.4 Load Testing

Targets:

- 100 concurrent users browsing archive
 - 20 concurrent users drafting and auto-saving
 - API response time p95 < 500ms under load
-

13. Deployment Checklist

Pre-Launch

- All 5 canonical texts uploaded with complete metadata and source passages
- 15–20 seed submissions published across all texts and both activity types
- Seed submissions include varied tones (hopeful, tragic, ironic, ambiguous)
- Admin account created and secured
- Database migrations run and verified
- Environment variables configured (JWT secret, database URL, admin list)
- SSL certificate active
- Rate limiting enabled
- Error tracking configured (Sentry or similar)
- Backup strategy defined (daily database snapshots)

Launch Day

- Deploy to production
- Verify all 5 text detail pages load correctly
- Test registration flow with fresh email
- Test complete creative loop: browse → register → draft → auto-save → publish → view in archive
- Verify archive search and filters
- Check mobile responsiveness on iOS Safari and Android Chrome
- Monitor error logs for 4 hours post-launch

Post-Launch Week 1

- Daily review of new registrations and submissions
- Respond to any user-reported issues within 24 hours
- Monitor auto-save failure rates
- Collect informal feedback from first 10 active users

14. Post-Launch Monitoring

Metrics Dashboard

Metric	Source	Alert Threshold
Daily active users	Application logs	<5 in first week
Registration-to-submission rate	Database query	<10%
Draft-to-publish rate	Database query	<30%
Average drafting time	Frontend events	<10 minutes or >2 hours
Auto-save failure rate	Backend logs	>5%
API error rate	Backend logs	>2%
Page load time (p95)	Frontend analytics	>3s

Weekly Review Questions

1. Which canonical text generates the most submissions?
2. What is the average length of published reimaginings?
3. Are users returning to write second submissions?
4. Which tags are most frequently used?
5. Are there any patterns in abandoned drafts (common drop-off points)?

Phase 1 Exit Criteria

Before beginning Phase 2, confirm:

- At least 50 registered users
- At least 20 published submissions
- Registration-to-submission rate >15%
- Draft-to-publish rate >40%
- No critical bugs unresolved for >48 hours

- User feedback indicates desire for peer review (validates Phase 2)

Appendix A: Activity Type Definitions (Phase 1 Only)

Alternative Ending

The user writes a new conclusion to the canonical text, diverging from the original at a specific point. The new ending must respect the tone, logic, and emotional shape of the source while exploring a different resolution.

Examples:

- Othello: What if Emilia exposed Iago before the murder?
- Jane Eyre: What if Jane never returned to Rochester?

Counterfactual Writing

The user explores how the narrative changes if a key event had never happened, a decision was reversed, or a confession was delayed or accelerated.

Examples:

- Macbeth: What if Macbeth refused the prophecy?
- Pride and Prejudice: What if Darcy never wrote the letter?

Appendix B: Glossary

Term	Definition
Canonical Text	A literary work in the public domain that serves as source material for reimagining
Draft	An in-progress, unpublished reimagining saved to the user’ s account
Submission	A published reimagining visible in the public archive
Activity Type	The category of creative transformation applied (Alternative Ending, Counterfactual Writing, etc.)
Author Note	A brief explanation (50–300 characters) of the user’ s creative intent
Source Passage	A curated excerpt (300–800 words) from the canonical text provided as reference
Seed Content	Pre-published submissions created before launch to populate the empty archive
Drafting Canvas	The specialized writing interface with side-by-side source text view